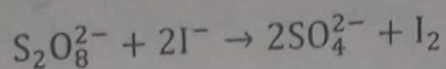


Experiment: Investigation of the Kinetics of the reaction**Theory:**

The rate of the above reaction is given by the expression;

$$\text{Rate} = \frac{[-d \text{S}_2\text{O}_8^{2-}]}{dt} = k[\text{S}_2\text{O}_8^{2-}]^n [\text{I}^-]^m$$

Using the following procedure, determine the value of **n** and **m**, the **rate constant**, **k** at 15°C, 20°C, 25°C, 30°C and 35°C and the activation energy of above reaction.

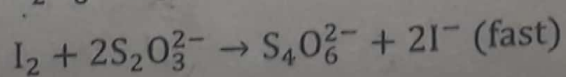
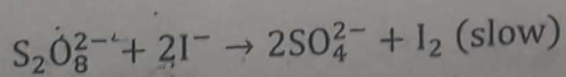
Procedure:

(a) To 5.00 cm³ of 1.00 mol dm⁻³ KI, 5.00 cm³ of 0.004 mol dm⁻³ Na₂S₂O₃ and drops of starch solution in a boiling tube at 25°C, add 10.00 cm³ of 0.02 mol dm⁻³ Na₂S₂O₈ and determine the time taken to blueness (t).

(b) To 5.00 cm³ of 0.004 mol dm⁻³ Na₂S₂O₃, 5.00 cm³ of 1.00 mol dm⁻³ KI, 5.00 cm³ of distilled water and 5 drops of starch solution in a boiling tube at 25°C, add 5.00 cm³ of 0.02 mol dm⁻³ Na₂S₂O₈ and determine the time to blueness (t).

(c) To 10.00 cm³ of 1.00 mol dm⁻³ KI, 5.00 cm³ of 0.004 mol dm⁻³ Na₂S₂O₃ and 5 drops of starch solution in a boiling tube at 25°C, add 5.00 cm³ of 0.02 mol dm⁻³ Na₂S₂O₈ and determine the time taken to blueness (t).

(d) Repeat part (a) at 15°C, 20°C, 30°C and 35°C



(e) Calculate the initial concentration of S₂O₈²⁻ and I⁻ in the mixture (a), (b) and (c).

(f) Calculate the initial rate of the reaction mixture (a), (b) and (c) at 25°C and mixture (a) at 15°C, 20°C, 30°C and 35°C using the relationship.

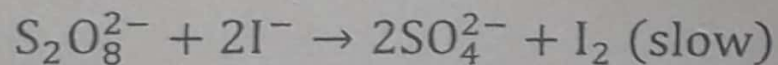
$$\text{Initial rate} = -(\overset{0}{C_t} - \bar{C}_0)/t$$

Where, C_0 and C_t are the concentrations of the $\text{S}_2\text{O}_8^{2-}$ at $t=0$ and $t=t$ respectively

(g) Using your results, calculate the value of n and express it to the nearest whole number.

(h) Using your results, calculate the value of m and express it to the nearest whole number.

(i) Using the results from experiment (a) and (d) calculate the **rate constant, k** of the reaction at 15°C, 20°C, 30°C and 35°C and hence determine the **activation energy E_a** the reaction,



(j) Comment on the reproducibility of your results.